THE NORMALIZATION OF A NAGATA ALGEBRAIC SPACE IS FINITE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement is one sentence; the proof is a short reduction to the construction of the normalization and the scheme case.

Editorial note. The three references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Nagata implies locally Noetherian for algebraic spaces. used at proof lines 10656-10657. Makes the definition of the normalization applicable.
- The construction of the normalization of an algebraic space. Lemma lemma-normalization and Definition definition-normalization, used at proof lines 10657-10659. Allows the reduction to the scheme case.
- The scheme case: the normalization of a Nagata scheme is finite. Morphisms, Lemma morphisms-lemma-nagata-normalization, used at proof lines 10659-10660. Is the input that closes the argument.

Source reference. Original definition 0BB2: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14230–14230. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 07U4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9959–9959. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 035S: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4139–4139. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X be a Nagata algebraic space over S. The normalization $\nu : X^\nu \to X$ is a finite morphism.*

Proof. Since X being Nagata is locally Noetherian, Definition 0BB2 applies. By construction of X^ν in Lemma 07U4 we immediately reduce to the case of schemes which is Morphisms, Lemma 035S. □

Date: stacks-project:spaces-morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.